

1. Which type of learning uses labeled data with known outputs?

- a) Supervised
- b) Unsupervised
- c) Reinforcement
- d) Self-supervised

Answer: a

2. Which algorithm is commonly used for binary classification?

- a) Linear Regression
- b) Logistic Regression
- c) K-Means
- d) Apriori

Answer: b

3. What does overfitting mean in model training?

- a) Model fails to learn training data
- b) Model performs well on training data but poorly on unseen data
- c) Model is too simple
- d) Model has no parameters

Answer: b

4. Which metric is used to evaluate classification models?

- a) R-squared
- b) F1 score
- c) Mean Absolute Error
- d) RMSE

Answer: b

5. Which dataset is commonly used to learn classification?

- a) Iris
- b) Titanic
- c) MNIST
- d) All of the above

Answer: d

6. What is the output of a linear regression model?

- a) A class label
- b) A continuous value
- c) A probability
- d) A cluster id

Answer: b

7. Which technique splits data into training and testing sets?

- a) Cross-validation
- b) Gradient descent
- c) Normalization
- d) Feature scaling

Answer: a

8. Which of the following is an unsupervised learning algorithm?

- a) K-Means
- b) Linear Regression
- c) Decision Tree
- d) Logistic Regression

Answer: a

9. What is the purpose of the learning rate in gradient descent?

- a) Controls step size
- b) Controls dataset size
- c) Controls number of features
- d) Controls model depth

Answer: a

10. Which library is most popular for machine learning in Python?

- a) NumPy
- b) Pandas
- c) Scikit-learn
- d) Requests

Answer: c

11. Explain the bias-variance tradeoff in machine learning models.

(Write your answer in the space provided.)

12. Describe the steps you would take to build a spam email classifier from scratch.

(Write your answer in the space provided.)